

Project Proposal: Multimodal Generative NPC Interaction System with Automated Persona Construction for VR/RPG Environments

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Problem Statement & Motivation

Traditional NPC interactions in RPG and VR environments are strictly limited to fixed dialogue trees, predefined branches, or menu-based selections. This scripted nature makes conversations repetitive and unnatural, which severely reduces player immersion, especially in VR settings. Beyond dialogue generation itself, a deeper and underexplored challenge is persona scalability: existing generative NPC systems (Park et al., 2023) rely on manually authored character descriptions, a process that does not scale to the hundreds of NPCs typical in RPG environments. Our goal is to address both problems by building a multimodal generative AI framework in Unity, allowing players to speak naturally to NPCs that are equipped with automatically generated, structured personas rather than hand-crafted prompts.

Proposed Method & Architecture

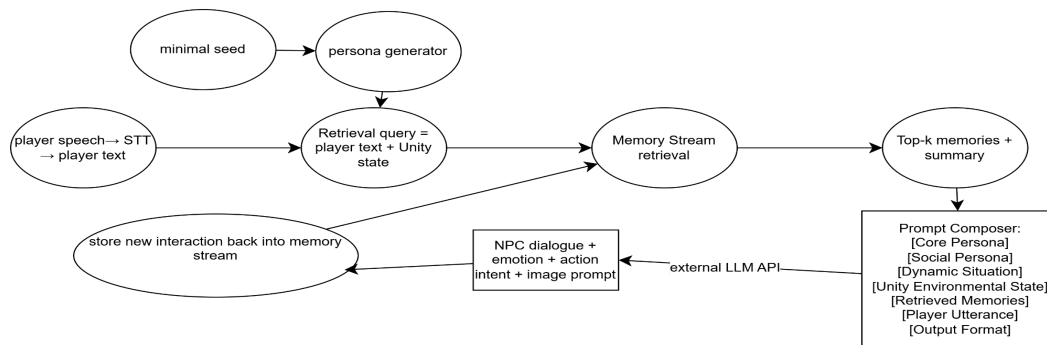
We will build a Unity-based VR prototype where the interaction is driven by a real-time, three-step generative pipeline. Drawing inspiration from the Generative Agents (Park et al., 2023) architecture, our system will operate as follows:

1. **Speech-to-Text (STT) Processing:** The system will capture the player's natural spoken audio and convert it into text using an open-source model (e.g., Whisper). If time permits, we plan to construct and fine-tune this STT module locally to optimize for VR latency.
2. **LLM-Based Adaptive Dialogue Generation with Automated Persona Construction:** The transcribed text is fed into a Python-based LLM backend acting as the NPC's dialogue architecture. A key contribution of our system is replacing the conventional hand-writing persona prompt with an automated three-layer persona generation pipeline. Given only a minimal seed specification (e.g., occupation, personality tags, and inter-character relationships), an LLM automatically generates: (a) a core persona layer, which stores stable identity information, long-term memory; (b) a social persona layer, which represents the NPC's relationships with other NPCs; and (c) a dynamic situation layer, which is updated during interaction and records the NPC's statements as a short-term memory. These structured layers will be packed into the external LLM API (if it is possible, we can train the language model by ourselves) with the environmental state. When the player speaks, the backend constructs a retrieval query based on the player's speech, the relevant NPC's memory stream, and the current Unity state. Furthermore, it ranks memories using semantic relevance, recency, and importance. Only the top relevant memories are summarized and inserted into

the final prompt, instead of inserting the entire memory stream. And then LLM outputs the response in the text, the actual decision, like a policy's role.

3. **Generative Text-to-Speech (TTS) & Visual Lip-Sync:** To maximize VR immersion, we have upgraded our initial text-to-image concept. The LLM's text output will be vocalized using state-of-the-art open-source generative TTS models. We will explore VITS and ChatTTS etc. This generated audio will then drive a real-time 3D facial animation/lip-sync module directly on the NPC avatar in Unity.

4. The entire workflow:



Dataset Specifications

Since our pipeline leverages pre-trained foundational models, we do not need to train an LLM from scratch. Instead, our datasets will serve three specific purposes:

- **Persona Seed Dataset:** We will author 20–30 minimal NPC seed specifications (occupation, personality tags, relationships) covering diverse RPG archetypes (e.g., blacksmith, court mage, merchant). These serve as inputs to our automated persona generation pipeline and as ground truth anchors for consistency evaluation.
- **Evaluation Dataset:** We will curate a new, custom test suite comprising 50–100 diverse RPG conversational prompts (e.g., quest inquiries, small talk, and adversarial tests designed to induce persona drift). We are building this locally because generic public text datasets fail to capture the highly specific context of immersive gaming required to test our system's logic.
- **Voice Reference Dataset:** To enable the Generative TTS module, we will utilize a small subset of high-quality public audio data (e.g., the VCTK corpus or public-domain game extractions). These 3–10 second clips will provide the necessary acoustic references for zero-shot voice cloning, giving our NPC a unique persona.

Evaluation Method

We will evaluate the system's performance using a mixed-methods approach:

- **Quantitative (System):** We will measure end-to-end system latency (STT -> LLM -> TTS), alongside objective lip-sync confidence scores.
- **Quantitative (Persona Consistency):** We evaluate generated personas using three automatic metrics from (Abdulhai et al., 2025): prompt-to-line consistency (does the response match the persona seed?), line-to-line consistency (does the NPC contradict itself across turns?), and using TIFA to rate the consistency (does the NPC give stable answers to repeated factual questions about itself?). We compare our automated persona against a hand-authored baseline.

References

1. Abdulhai, M., Cheng, R., Clay, D., Althoff, T., Levine, S., & Jaques, N. (2025). *Consistently Simulating Human Personas with Multi-Turn Reinforcement Learning*. arXiv preprint arXiv:2511.00222.
2. Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). *Generative Agents: Interactive Simulacra of Human Behavior*. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*.

